Pre-Analysis Plan – Hypothesis 1

Hypothesis Statement

Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables

Dependent Variable:

- plh0308_v2: Level of agreement with the statement "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally."
    - Missing values (originally coded with negative values) will be recoded as “.”

Independent Variable:

- d_daughter: Dummy variable taking value 1 if the respondent has at least one daughter born before the survey year, 0 otherwise
- Construction: Created by first identifying the year when the first daughter was born (FDY) across all birth variables (kidgeb01 through kidgeb19 where corresponding kidsex variables equal 2 for female), then creating the dummy based on whether FDY < syear.

Control Variables:

- sex: Participant gender (used as categorical with i.sex)
- germborn: Dummy for German-born participants (used as categorical with i.germborn)
- hghinc: Net monthly household income (missing values coded as .)
- gebjahr: Birth year of participant (used as fixed effects with i.gebjahr)
- sumkids_dynamic: Number of children born before the survey year (dynamically calculated based on birth years relative to survey year)

Inclusion/Exclusion Criteria

Inclusion Criteria:

- Participants from private households only (inlist(pop,1,2))
- Respondents who had their first child before 2018 (kidgeb01<2018 & kidgeb01>0)

Time Period:

- Survey data from 2018 (when the dependent variable was asked)

Exact Analysis

Regression Specification:

reg plh0308_v2 d_daughter i.sex i.germborn hghinc i.gebjahr sumkids_dynamic, vce(robust)

Model Details:

- Dependent variable: plh0308_v2
- Key independent variable: d_daughter
- Control variables:
    - i.sex (participant gender fixed effects)
    - i.germborn (German-born status fixed effects)
    - hghinc (household net income, continuous)
    - i.gebjahr (birth year fixed effects)
    - sumkids_dynamic (number of children, continuous)
- Standard errors: Robust standard errors (vce(robust))
- Estimation method: Ordinary Least Squares (OLS) linear regression

Comment:
Our measure of support for gender equality is based on respondents' level of agreement with the statement: “It’s best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally.” Accordingly, we estimate a linear OLS regression where the dependent variable is plh0308_v2, and the key independent variable is a dummy, d_daughter, which takes the value 1 if the respondent had at least one daughter at the time of responding (survey year: 2018).

Our objective is to estimate the causal effect of having a daughter on gender equality attitudes. While the gender of a child is generally considered to be randomly assigned, fertility decisions are not. Therefore, we restrict our sample to respondents who were already parents at the time of the survey.

However, it is important to acknowledge that subsequent fertility decisions may be influenced by the gender of previously born children, i.e. "gender stopping rules" (Angrist and Evans, 1998). If such decisions are driven by gender preferences, this could threaten the causal interpretation of our estimates. We tested the possibility of gender stopping rules by regressing the number of children on the gender of the first born. By doing so, we found no evidence of correlation (coef 0.008833, p-value=0.848), indicating that gender stopping rules are not extended among the participants. Additionally, to mitigate this concern, we control for the number of children the respondent had at the time of completing the questionnaire.

We include the following controls: participant gender, net monthly income of the household, a dummy for German born participants, and birth year of the participant fixed effects.

Interpretation of Results

Test Statistic:

- The coefficient estimate on d_daughter
- t-statistic and corresponding p-value using robust standard errors

Hypothesis Testing:

- Null hypothesis: Having a daughter has no effect on gender equality attitudes (coefficient = 0)
- Alternative hypothesis: Having a daughter increases support for gender equality (coefficient > 0)

Interpretation Criteria:

- In line with hypothesis: If the coefficient on d_daughter is positive (> 0), indicating that having a daughter increases support for gender equality
- Not in line with hypothesis: If the coefficient on d_daughter is negative or zero (≤ 0)

Pre-Analysis Plan – Hypothesis 2

Hypothesis Statement

Having daughters leads to more left-leaning political preferences.

Variables

Dependent Variable:

- plh0004**:** Self-placement on the left-right political spectrum, using a scale from 0 ("ganz links" / far left) to 10 ("ganz rechts" / far right)
    - Missing values (originally coded with negative values) will be recoded as “.”

Independent Variable:

- d_daughter: Dummy variable taking value 1 if the respondent has at least one daughter born before the survey year, 0 otherwise
- Construction: Created by first identifying the year when the first daughter was born (FDY) across all birth variables (kidgeb01 through kidgeb19 where corresponding kidsex variables equal 2 for female), then creating the dummy based on whether FDY < syear.

Control Variables:

- sex: Participant gender (used as categorical with i.sex)
- germborn: Dummy for German-born participants (used as categorical with i.germborn)
- hghinc: Net monthly household income (missing values coded as .)
- gebjahr: Birth year of participant (used as fixed effects with i.gebjahr)
- sumkids_dynamic: Number of children born before the survey year (dynamically calculated based on birth years relative to survey year)

Inclusion/Exclusion Criteria

Inclusion Criteria:

- Participants from private households only (inlist(pop,1,2))
- Respondents who had their first child before 2019 (kidgeb01<2019 & kidgeb01>0)
- Survey year 2019 only (syear==2019)

Time Period:

- Survey data from 2019 (when the dependent variable was asked)

Exact Analysis

Regression Specification:

reg plh0004 d_daughter i.sex i.germborn hghinc i.gebjahr sumkids_dynamic, vce(robust)

Model Details:

- Dependent variable: plh0004
- Key independent variable: d_daughter
- Control variables:
    - i.sex (participant gender fixed effects)
    - i.germborn (German-born status fixed effects)
    - hghinc (household net income, continuous)
    - i.gebjahr (birth year fixed effects)
    - sumkids_dynamic (number of children, continuous)
- Standard errors: Robust standard errors (vce(robust))
- Estimation method: Ordinary Least Squares (OLS) linear regression

Comment:
Our measure political preferences is based on respondents' self-placement on the left–right political spectrum, using a scale from 0 ("ganz links" / far left) to 10 ("ganz rechts" / far right). Accordingly, we estimate a linear OLS regression where the dependent variable is plh0004, and the key independent variable is a dummy, d_daughter, which

takes the value 1 if the respondent had at least one daughter at the time of responding (survey year: 2019).

Our objective is to estimate the causal effect of having a daughter on political preferences. While the gender of a child is generally considered to be randomly assigned, fertility decisions are not. Therefore, we restrict our sample to respondents who were already parents at the time of the survey.

However, it is important to acknowledge that subsequent fertility decisions may be influenced by the gender of previously born children, i.e. "gender stopping rules" (Angrist and Evans, 1998). If such decisions are driven by gender preferences, this could threaten the causal interpretation of our estimates. We tested the possibility of gender stopping rules by regressing the number of children on the gender of the first born. By doing so, we found no evidence of correlation (coef 0.008833, p-value=0.848), indicating that gender stopping rules are not extended among the participants. Additionally, to mitigate this concern, we control for the number of children the respondent had at the time of completing the questionnaire.

We include the following controls: participant gender, net monthly income of the household, a dummy for german born participants, and birth year of the participant fixed effects.

Interpretation of Results

Test Statistic:

- The coefficient estimate on d_daughter
- t-statistic and corresponding p-value using robust standard errors

Hypothesis Testing:

- Null hypothesis: Having a daughter has no effect on political preferences (coefficient = 0)
- Alternative hypothesis: Having a daughter leads to more left-leaning political preferences (coefficient < 0)

Interpretation Criteria:

- In line with hypothesis: If the coefficient on d_daughter is negative (< 0), indicating that having a daughter shifts political preferences toward the left (lower values on the 0-10 scale)
- Not in line with hypothesis: If the coefficient on d_daughter is positive or zero (≥ 0)

Pre-Analysis Plan – Hypothesis 3

Hypothesis Statement

Having daughters leads to more gender equality in the household (as measured by behaviours or outcomes).

Variables

Dependent Variable:

- house_workload: Total hours spent on housework, childcare, and repairs (continuous variable)
    - Construction: Sum of hours across multiple activities:
        - Housework: pli0012_h (Hours Saturday Housework) + pli0016_h (Hours Sunday Housework) + pli0043_h (Hours Weekday Housework)
        - Childcare: pli0019_h (Hours Saturday Child Care) + pli0022_h (Hours Sunday Child Care) + pli0044_h (Hours Weekday Child Care)
        - Repairs: pli0049_h (Hours Weekday Repairs) + pli0031_h (Hours Saturday Repairs)
    - Missing values: Missing values (originally coded with negative values) will be recoded as "."
    - Note: pli0177 (Hours Sunday Repairs) is excluded as it is missing for syear==2019

Independent Variable:

- d_daughter: Interaction term (female × having_daughter) capturing the change in gender gap in households workload when having daughters
    - Construction: d_daughter = female × having_daughter
    - Where:
    - female: Dummy variable (1 if sex==2, 0 otherwise)
    - having_daughter: Dummy variable taking value 1 if the respondent has at least one daughter born before the survey year, 0 otherwise

    - having_daughter construction: Created by first identifying the year when the first daughter was born (FDY) across all birth variables (kidgeb01 through kidgeb19 where corresponding kidsex variables equal 2 for female), then creating the dummy based on whether FDY < syear

Control Variables:

- having_daughter: Main effect of having a daughter (dummy variable)
- female: Main effect of participant gender (dummy variable)
- germborn: Dummy for German-born participants (used as categorical with i.germborn)
- hghinc: Net monthly household income (missing values coded as .)
- gebjahr: Birth year of participant (used as fixed effects with i.gebjahr)
- sumkids_dynamic: Number of children born before the survey year (dynamically calculated based on birth years relative to survey year)

Inclusion/Exclusion Criteria

Inclusion Criteria:

- Participants from private households only (inlist(pop,1,2))
- Respondents who had their first child before 2019 (kidgeb01<2019 & kidgeb01>0)
- Survey year 2019 only (syear==2019)

Time Period:

- Survey data from 2019 (when the dependent variable was asked)

Exact Analysis

Regression Specification:

reg house_workload d_daughter having_daughter female i.germborn hghinc i.gebjahr sumkids_dynamic, vce(robust)

Model Details:

- Dependent variable: house_workload
- Key independent variable: d_daughter
- Control variables:

    - having_daughter (main effect of having daughters)
    - female (main effect of participant gender)
    - i.germborn (German-born status fixed effects)
    - hghinc (household net income, continuous)
    - i.gebjahr (birth year fixed effects)
    - sumkids_dynamic (number of children, continuous)
- Standard errors: Robust standard errors (vce(robust))
- Estimation method: Ordinary Least Squares (OLS) linear regression

Comment:
Our measure of gender equality in the household is based on gender gaps in time devoted in housework, childcare and repairs. We created a variable house_workload aggregating all time devoted to these activities. We estimate a linear OLS regression where the dependent variable is house_workload, and the key independent variable is the interaction between the gender of the participant (female) and having_daughters (which takes the value 1 if the respondent had at least one daughter at the time of responding, i.e. year 2019). The coefficient of the uninteracted "female" variable captures gender gaps in house_workload, while the coefficient of the interaction captures how the gender gap in house_workload is affected by having daughters.

Our objective is to estimate the causal effect of having a daughter on gender equality in the household. While the gender of a child is generally considered to be randomly assigned, fertility decisions are not. Therefore, we restrict our sample to respondents who were already parents at the time of the survey.

However, it is important to acknowledge that subsequent fertility decisions may be influenced by the gender of previously born children, i.e. "gender stopping rules" (Angrist and Evans, 1998). If such decisions are driven by gender preferences, this could threaten the causal interpretation of our estimates. We tested the possibility of gender stopping rules by regressing the number of children on the gender of the first born. By doing so, we found no evidence of correlation (coef 0.008833, p-value=0.848), indicating that gender stopping rules are not extended among the participants. Additionally, to mitigate this concern, we control for the number of children the respondent had at the time of completing the questionnaire.

We include the following controls: participant gender, net monthly income of the household, a dummy for German born participants, and birth year of the participant fixed effects.

Interpretation of Results

Test Statistic:

- The coefficient estimate on d_daughter
- t-statistic and corresponding p-value using robust standard errors

Hypothesis Testing:

- Null hypothesis: Having a daughter has no effect on household gender equality (coefficient = 0)
- Alternative hypothesis: Having a daughter leads to more gender equality in the household (coefficient < 0)

Interpretation Criteria:

- In line with hypothesis: If the coefficient on d_daughter is negative (< 0), indicating that having a daughter reduces the gender gap in household workload (i.e., women do relatively less household work or men do relatively more when there are daughters in the household)
- Not in line with hypothesis: If the coefficient on d_daughter is positive or zero (≥ 0), indicating no improvement or worsening of gender equality in household workload

Pre-Analysis Plan – Hypothesis 4

Hypothesis Statement

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables

Dependent Variable:

- plh0308_v2: Level of agreement with the statement "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally."
    - Missing values (originally coded with negative values) will be recoded as "." Independent Variable:
- **d_daughter**: Interaction term (female × having_daughter)
    - This captures the differential effect of having a daughter for women compared to men
    - Construction: d_daughter = female × having_daughter, where:
        - female = 1 if respondent is female (sex==2), 0 if male
        - having_daughter = 1 if respondent has at least one daughter born before survey year, 0 otherwise
- **having_daughter**: Dummy variable for having at least one daughter
    - Construction: Created by first identifying the year when the first daughter was born (FDY) across all birth variables (kidgeb01 through kidgeb19 where corresponding kidsex variables equal 2 for female), then creating the dummy based on whether FDY < syear
- **female**: Gender dummy variable
    - female = 1 if sex==2 (female), 0 if sex==1 (male)

Control Variables:

- germborn: Dummy for German-born participants (used as categorical with i.germborn)
- hghinc: Net monthly household income (missing values coded as .)

- gebjahr: Birth year of participant (used as fixed effects with i.gebjahr)
- sumkids_dynamic: Number of children born before the survey year (dynamically calculated based on birth years relative to survey year)

Inclusion/Exclusion Criteria

Inclusion Criteria:

- Participants from private households only (inlist(pop,1,2))
- Respondents who had their first child before 2018 (kidgeb01<2018 & kidgeb01>0)

Time Period:

- Survey data from 2018 (when the dependent variable was asked)

Exact Analysis

Regression Specification:

reg plh0308_v2 d_daughter having_daughter female i.germborn hghinc i.gebjahr sumkids_dynamic, vce(robust)

Model Details:

- Dependent variable: plh0308_v2
- Key independent variables:
    - d_daughter (interaction term: female × having_daughter)
    - having_daughter (main effect of having daughter)
    - female (main effect of being female)
- Control variables:
    - i.germborn (German-born status fixed effects)
    - hghinc (household net income, continuous)
    - i.gebjahr (birth year fixed effects)
    - sumkids_dynamic (number of children, continuous)
- Standard errors: Robust standard errors (vce(robust))
- Estimation method: Ordinary Least Squares (OLS) linear regression

Comment:
We replicate the same analysis for H1 with the only difference that we interact the explanatory variable having_daughter with a female dummy (taking value 1 if the survey respondent is female). The coefficient of the interaction will inform if the effect of having_daughter is different for fathers and mothers.

Interpretation of Results

Test Statistic:

- The coefficient estimate on d_daughter
- t-statistic and corresponding p-value using robust standard errors

Hypothesis Testing:

- Null hypothesis: The effect of having daughters on gender equality attitudes is the same for men and women (coefficient = 0)
- Alternative hypothesis: Having daughters has a larger positive effect for men than for women (coefficient < 0)

Interpretation Criteria:

- In line with hypothesis: If the coefficient on d_daughter is negative (< 0), indicating that the positive effect of having a daughter is smaller for women than for men.
- Not in line with hypothesis: If the coefficient on d_daughter is negative or zero (≤ 0)